Reservoir, spin and polarization dynamics of a polariton condensate

Korostelev Nikita

Aim of the study

The previous report, *Dynamics and spatial structure of exciton-polariton condensates under nonresonant pumping*, used a scalar condensate and one reservoir. Here we add an inactive reservoir, two circular components, constant coherent coupling, and momentum-dependent TE–TM splitting.

We examine reservoir transients, the response to pump polarization, and changes in phase dynamics and spatial polarization. Homogeneous calculations are followed by Gaussian-spot and annular pumping; Stokes parameters then provide a common description of the polarization states.

The condensation parameters are retained. Additional coefficients and their literature basis are given in Appendix A; they are not recalibrated using the earlier dispersion report.

1 Active and inactive reservoirs

1.1 Extension of the scalar model

In the previous model, the condensate field ψ and reservoir density n_R obeyed

$$\begin{aligned} i\hbar\partial_t\psi &= \left[-K\nabla^2 + g_C|\psi|^2 + g_R(n_R+\eta P_A) + \frac{i\hbar}{2}(Rn_R-\gamma_C)\right]\psi, \\ \partial_t n_R &= P_A - (\gamma_R + R|\psi|^2)n_R, \qquad K=\frac{\hbar^2}{2m}. \end{aligned} \tag{1}$$

Here, P_A replaces the previous pump notation P, distinguishing reservoir injection from the external pump introduced below.

The instantaneous energy shift $g_R\eta P_A$ is replaced by a dynamical inactive reservoir n_I. The external pump P_{in} populates this reservoir, which supplies the active reservoir n_A at rate Wn_I. Stimulated scattering from n_A feeds the condensate:

$$\begin{aligned} i\hbar\partial_t\psi &= \left[-K\nabla^2 + g_C|\psi|^2 + g_An_A + g_In_I + \frac{i\hbar}{2}(Rn_A-\gamma_C)\right]\psi, \\ \partial_t n_A &= Wn_I - (\gamma_A + R|\psi|^2)n_A, \\ \partial_t n_I &= P_{\text{in}} - (\gamma_I + W)n_I. \end{aligned} \tag{2}$$

Both reservoirs shift the condensate energy; $g_R\eta P_A$ is not added again. The model structures follow Refs. [1, 2].

To match the models, we choose

$$g_A = g_R, \qquad \gamma_A = \gamma_R, \qquad P_{\text{in}} = \frac{\gamma_I + W}{W}P_A, \qquad \frac{g_I}{W} = g_R\eta. \tag{3}$$

Under constant pumping, $n_I^* = P_A/W$. The particle supply and additional energy shift therefore reproduce their previous values:

$$Wn_I^* = P_A, \qquad g_In_I^* = g_R\eta P_A.$$

The stationary solutions of the scalar model are thus preserved. In the homogeneous case, the condensation threshold is also unchanged:

$$P_{\text{th}} = \frac{\gamma_C\gamma_A}{R} = 0.43333\ \mu\text{m}^{-2}\text{ps}^{-1}.$$

This matching does not require the transient dynamics to coincide.

1.2 Response to pump turn-on and turn-off

Define the particle transfer rate into the active reservoir as $Q_A = Wn_I$. Equations (2) and (3) give

$$\tau_I \frac{dQ_A}{dt} = P_A - Q_A, \qquad \tau_I = \frac{1}{\gamma_I + W}. \tag{4}$$

The supply to the active reservoir now responds on a finite time scale. For the main parameter set, $\tau_I = 9.63$ ps.

The comparison uses $P_A = 1.5P_{\mathrm{th}}$, initially empty reservoirs, and the same weak condensate seed. Both models approach

$$n_C^* = \frac{P_A - P_{\mathrm{th}}}{\gamma_C} = 1.19048\ \mu\mathrm{m}^{-2}.$$

Condensate growth starts later in the model with two reservoirs. Half of the steady-state density is first reached at 159.1 ps in the previous model and at 188.0 ps in the extended model. This delay reflects the entire transient and is not equal to τ_I.

After turn-off, the inactive reservoir continues to feed the active reservoir and slows the condensate decay (Fig. 1). The single-reservoir model also retains particles in its active reservoir, but its external supply stops immediately.

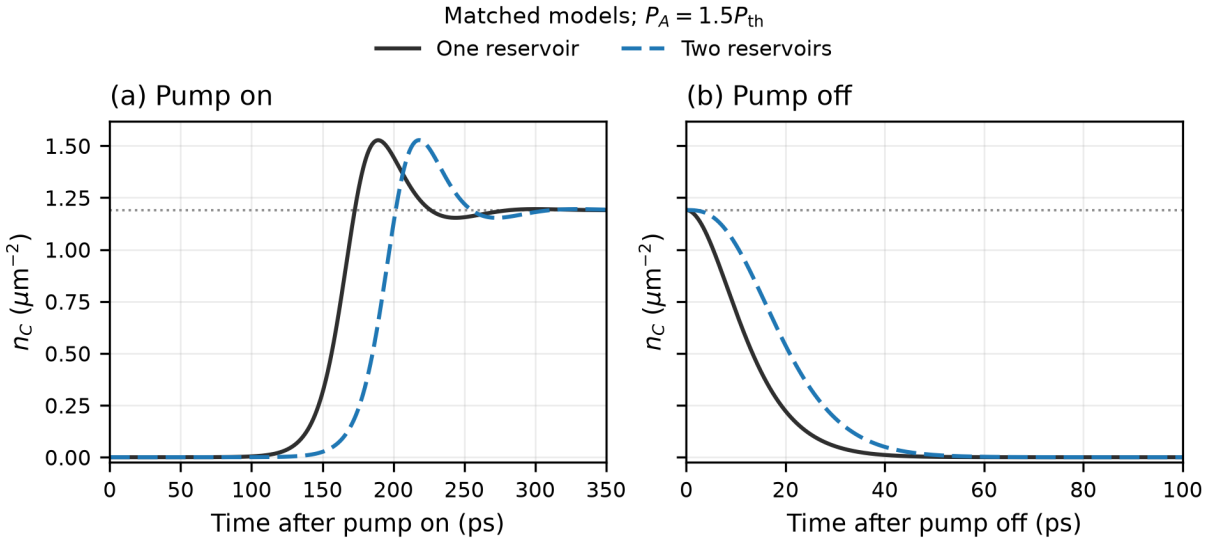


Figure 1: Condensate density after pump turn-on (a) and turn-off (b) for the single-reservoir and matched two-reservoir models. The on-state pump is $P_A = 1.5P_{\mathrm{th}}$; both runs use the same seed and initially empty reservoirs. The dotted line marks the common steady density. Time is measured from each switching event; turn-off occurs at 1000 ps.

2 Two spin components of the condensate

2.1 Model equations

The circular fields $\psi_\pm$ describe the two optically active angular-momentum projections along the cavity growth axis, not orbital circulation in the cavity plane. Their densities are $n_\pm = |\psi_\pm|^2$; each has an active and an inactive reservoir.

Before adding TE–TM splitting, we use

$$\begin{aligned} i\hbar\partial_t\psi_\sigma &= \left[-K\nabla^2 + V_\sigma + \frac{i\hbar}{2}(Rn_{A,\sigma} - \gamma_C)\right]\psi_\sigma - J\psi_{-\sigma}, \\ V_\sigma &= \alpha_1 n_\sigma + \alpha_2 n_{-\sigma} + g_A n_{A,\sigma} + g_I n_{I,\sigma}, \\ \partial_t n_{A,\sigma} &= W n_{I,\sigma} - (\gamma_A + Rn_\sigma)n_{A,\sigma} + \Gamma_{s,A}(n_{A,-\sigma} - n_{A,\sigma}), \\ \partial_t n_{I,\sigma} &= P_{\mathrm{in},\sigma} - (\gamma_I + W)n_{I,\sigma} + \Gamma_{s,I}(n_{I,-\sigma} - n_{I,\sigma}), \qquad \sigma = \pm. \end{aligned} \tag{5}$$

The coefficients α_1 and α_2 give interaction energy shifts for equal and opposite circular polarizations. Reservoir spin relaxation, proportional to Γ_s, transfers particles between reservoir components while conserving their sum [3]. Cross-spin reservoir interaction coefficients are zero.

Here $J = 0$. The diagonal term $\alpha_2 n_{-\sigma}$ changes phase evolution without directly transferring condensate particles; coherent transfer through J is considered in Chapter 3.

All density-dependent terms are local. Distinct spatial points are coupled by $-K\nabla^2$; reservoir diffusion and nonlocal density integrals are absent.

2.2 Density dependence on pump polarization

For the homogeneous problem, we set

$$P_{A,\pm} = sP_{\mathrm{th}}(1 \pm \epsilon), \qquad P_{\mathrm{in},\pm} = \frac{\gamma_I + W}{W}P_{A,\pm}. \tag{6}$$

The rescaled pump parameters $P_{A,\pm}$ differ from the actual transfers $Q_{A,\pm} = Wn_{I,\pm}$ when the inactive reservoir undergoes spin relaxation. Varying ϵ preserves the total pump $2sP_{\mathrm{th}}$; we use $s = 2$.

For $J = 0$, if both components have nonzero steady-state densities, gain-loss balance gives $n^*_{A,+} = n^*_{A,-} = \gamma_C/R$. Writing

$$D_I = \frac{\gamma_I + W}{\gamma_I + W + 2\Gamma_{s,I}},$$

we obtain

$$n^*_{I,\pm} = \frac{sP_{\mathrm{th}}}{W}(1 \pm \epsilon D_I), \qquad n^*_\pm = \frac{\gamma_A}{R}\left[s(1 \pm \epsilon D_I) - 1\right]. \tag{7}$$

These expressions apply when both predicted densities are positive, that is, $s(1-|\epsilon|D_I) > 1$. For the chosen parameters, $D_I = 0.34180$, and this condition holds throughout $-1 \le \epsilon \le 1$.

The normalized condensate imbalance, or degree of circular polarization, is

$$z = \frac{n_+ - n_-}{n_+ + n_-} = \frac{s\epsilon D_I}{s - 1}. \tag{8}$$

For $s = 2$, $z = 0.68360\epsilon$. Inactive-reservoir spin relaxation reduces the supply imbalance. Equal active-reservoir densities can therefore support unequal condensate densities. Integration reproduces Eq. (7) (Fig. 2): at $\epsilon = 0.3$, $n^*_+ = 2.86924$, $n^*_- = 1.89267\ \mu\mathrm{m}^{-2}$, and $z = 0.20508$.

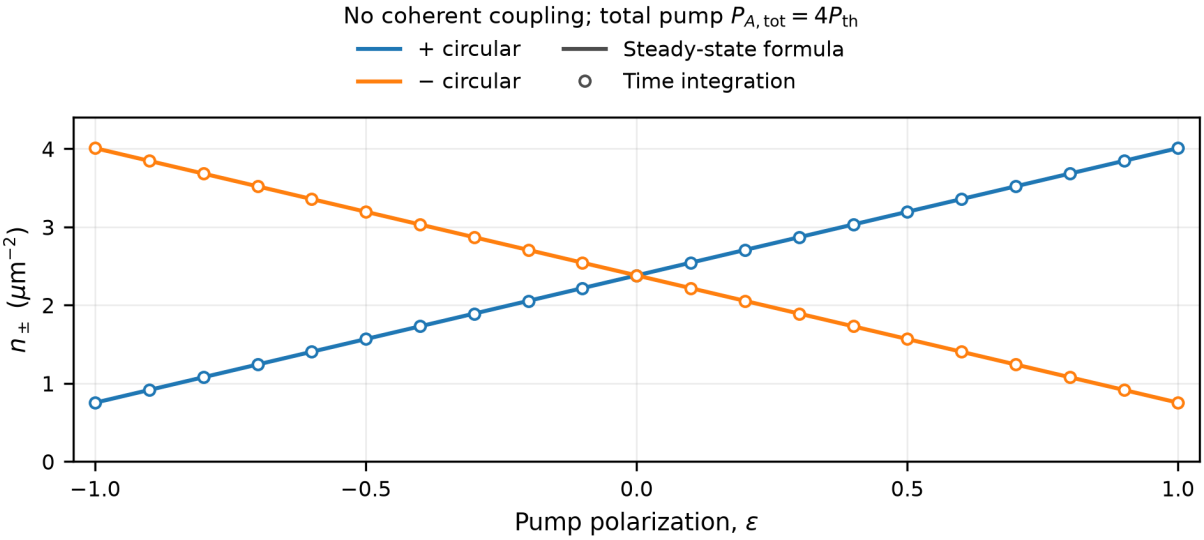


Figure 2: Homogeneous steady densities versus pump polarization ϵ, at total pump $4P_{\mathrm{th}}$ and $J = 0$, including reservoir spin relaxation. Lines show Eq. (7); circles show independent integrations. The component densities change while their sum remains constant.

2.3 Spatial distributions under spot and annular pumping

The spatial calculations resolve the cavity plane, $\mathbf{r} = (x, y)$. We use

$$P_{A,\pm}(r) = pP_{\mathrm{th}}(1 \pm \epsilon)f(r), \qquad f_{\mathrm{spot}}(r) = e^{-r^2/d^2}, \qquad f_{\mathrm{ring}}(r) = e^{-(r-r_0)^2/d^2}. \tag{9}$$

We use $d = 2.5$ μm, $r_0 = 10$ μm, and $p = 13.8$ for the spot or 3.3 for the ring. Each geometry retains its pump when couplings are compared. These working points lie above the estimated spatial onsets; P_{th} itself remains the homogeneous reference threshold.

Both components start with equal amplitudes and phase, and empty reservoirs. Annular runs share a weak prescribed non-axisymmetric seed perturbation (Appendix B). The main spatial comparisons use 512^2 points, $L = 256$ μm, and $\Delta t = 0.025$ ps. The separate high-J scan uses 256^2, as specified in Section 4.4. This preparation differs from the earlier report's annular series, so the old and new densities do not isolate a single changed coefficient.

Figure 3 shows the two component densities at $\epsilon = 0.3$, $J = \beta = 0$. The spot produces a central density maximum, whereas the annular pump produces an annular density distribution. The component populations differ, while their spatial profiles remain similar. We characterize them by

$$N_\pm = \int n_\pm(\mathbf{r})\, d^2r, \qquad Z = \frac{N_+ - N_-}{N_+ + N_-}. \tag{10}$$

Equation (8) does not directly determine the integrated Z, because spatial transport and reservoir inhomogeneity are present. The maps show the pump-induced imbalance; they do not isolate α_2. Setting $\alpha_2 = 0$ gives visually close profiles, but a precise small difference requires a dedicated error estimate.

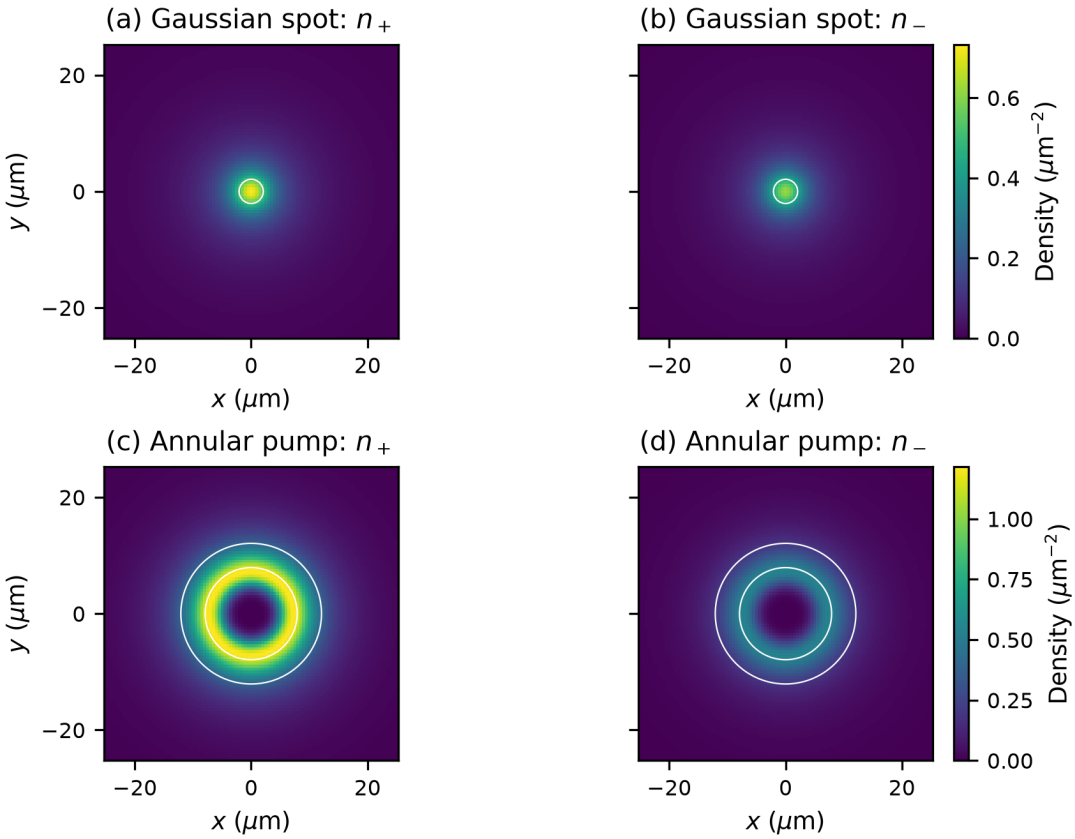


Figure 3: Circular-component densities under Gaussian-spot (a,b) and annular (c,d) pumping, with $\epsilon = 0.3$, $J = \beta = 0$, and $p = 13.8$ and 3.3, respectively. The 512^2 maps show late-time density profiles; each component pair shares an absolute color scale. White contours mark half the maximum pump intensity. Sampling details are listed in Table 1.

3 Internal Josephson dynamics in the 0D model

We now omit spatial derivatives in Eq. (5). Figures 4–6 describe independent homogeneous calculations, not a spatial average of Fig. 3.

3.1 Coherent coupling and particle exchange

The term $-J\psi_{-\sigma}$ coherently couples the circular components of one condensate. For equal diagonal energies, its linearly polarized eigenstates are separated by $2J$, independently of momentum. Internal Josephson dynamics therefore require no TE–TM splitting. Nonresonant pumping supplies reservoir particles, while the field equations determine the relative phase [4].

Writing $\psi_\pm = \sqrt{n_\pm}e^{i\varphi_\pm}$, define

$$\theta = \varphi_+ - \varphi_-, \qquad I_J = \frac{2J}{\hbar}\,\mathrm{Im}(\psi_-^*\psi_+) = \frac{2J}{\hbar}\sqrt{n_+ n_-}\,\sin\theta. \tag{11}$$

Positive I_J denotes transfer into the $+$ component:

$$\dot{n}_+ = (Rn_{A,+} - \gamma_C)n_+ + I_J, \qquad \dot{n}_- = (Rn_{A,-} - \gamma_C)n_- - I_J. \tag{12}$$

The relative phase obeys

$$\dot{\theta} = -\frac{V_+ - V_-}{\hbar} - \frac{2J}{\hbar}\frac{z}{\sqrt{1-z^2}}\cos\theta. \tag{13}$$

A fixed relative phase need not be zero: both field phases may evolve at a common rate. Spatial uniformity is assumed in this 0D model, not established by phase locking.

3.2 Three observed dynamical regimes

The calculations use $s = 2$, $\epsilon = 0.3$, and identical initial conditions. Three examples are retained from the previous version:

- At $J = 5$ μeV, the densities oscillate periodically and the relative phase continues to wind.
- At $J = 6.5$ μeV, density oscillations persist, but the relative phase oscillates within a bounded interval.
- At $J = 10$ μeV, the densities and relative phase become constant.

Figures 4 and 5 show the densities and phase over the same late-time interval. Vanishing mean phase drift does not imply that the density oscillations have ceased.

At $J = 10$ μeV, the steady current is $I_J \simeq 0.01477$ $\mu\mathrm{m}^{-2}\mathrm{ps}^{-1}$. It compensates the unequal gain-loss balance of the two components. A fixed relative phase can therefore coexist with nonzero particle transfer, while the fields rotate at a common frequency.

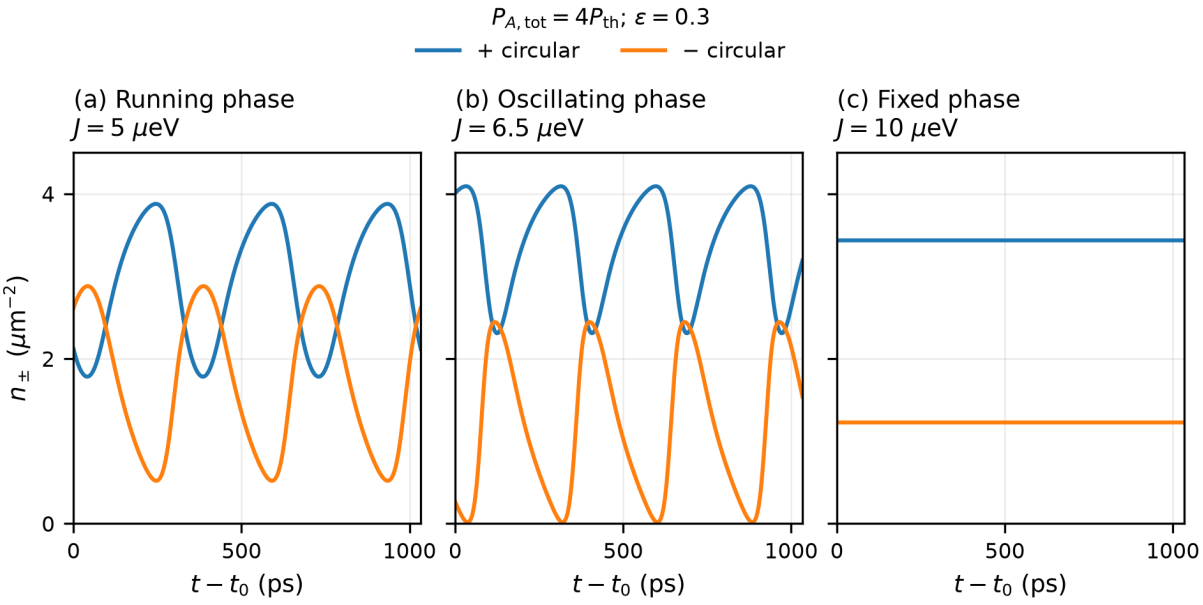


Figure 4: Homogeneous densities at $J = 5$, 6.5, and 10 μeV: running phase (a), bounded phase oscillations (b), and fixed phase (c). The total pump is $4P_{\mathrm{th}}$, with $\epsilon = 0.3$. All panels share component colors and a density scale. The interval is $16966.5 \leq t \leq 18000$ ps, displayed relative to its start.

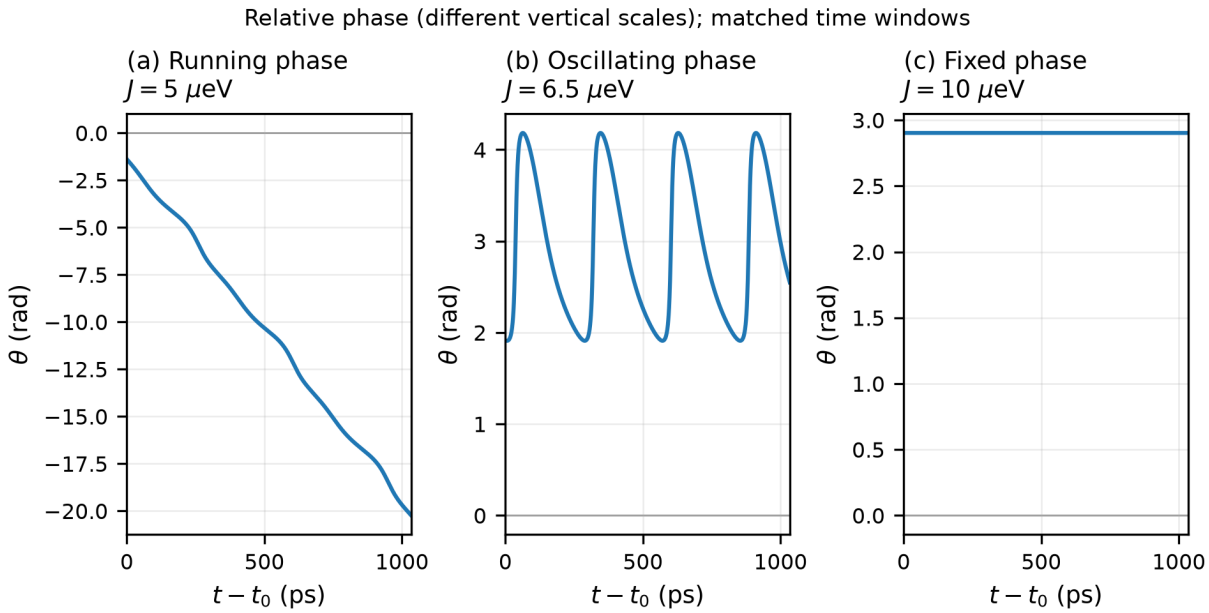


Figure 5: Unwrapped relative phase $\theta = \arg(\psi_+\psi_-^*)$ for the calculations and interval of Fig. 4. The vertical scales differ. Only integer multiples of 2π are subtracted, preserving the steady state's nonzero phase difference.

3.3 Dependence on the coupling strength

Two quantities distinguish the regimes:

$$A_z = \frac{z_{\max} - z_{\min}}{2}, \qquad \Omega_\theta = \langle \dot{\theta} \rangle. \tag{14}$$

Here A_z is the imbalance half-amplitude. Figure 6 plots the mean relative-phase drift $\Omega_\theta = \langle \dot{\theta} \rangle$, evaluated over complete late-time cycles. With $\theta = \arg(\psi_+\psi_-^*)$, the running-phase examples have negative drift. This is the opposite sign to the frequency difference $\overline{\omega_+ - \omega_-}$ when $\omega_\sigma = -\partial_t \arg \psi_\sigma$.

In the retained homogeneous series, the mean phase drift vanishes near $J = 6.4\ \mu$eV, while density oscillations persist to stronger coupling. Their disappearance is bracketed by the sampled values 6.84 and 6.86 μeV. At 6.38 and 6.40 μeV, one component becomes too weak during part of the cycle for its phase to be determined reliably. Density periodicity is confirmed, but no phase-drift value is assigned.

These transitions concern an existing condensate, not its condensation threshold. Independent runs with the same initial state sample one preparation protocol, rather than all possible branches.

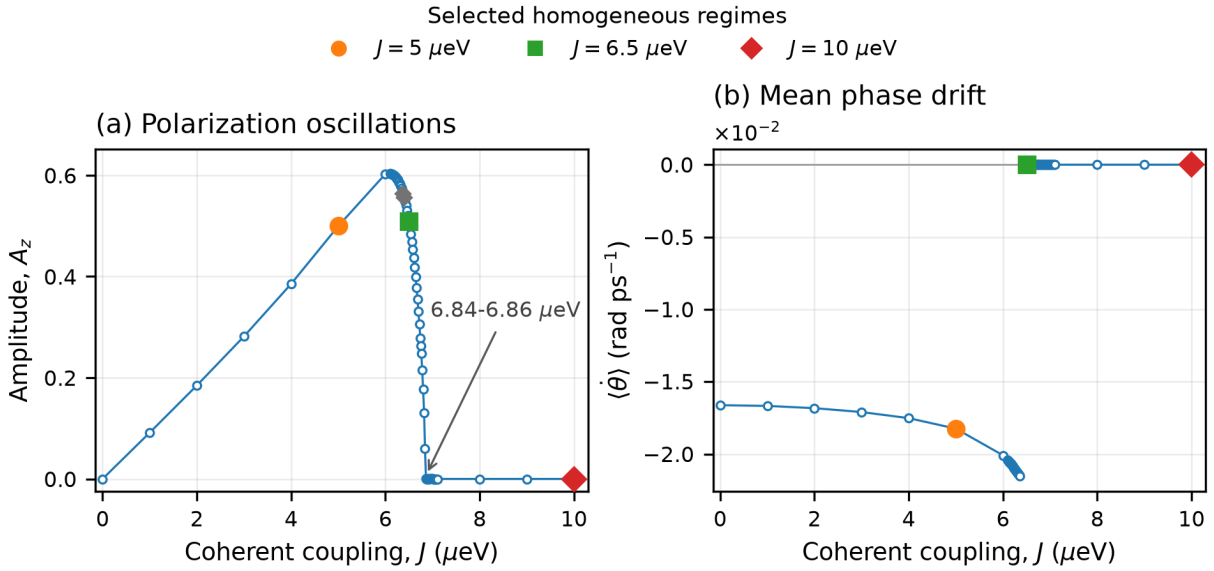


Figure 6: Homogeneous coupling scan at total pump $4P_{\mathrm{th}}$ and $\epsilon = 0.3$: imbalance half-amplitude A_z (a) and mean relative-phase drift $\Omega_\theta = \langle\dot{\theta}\rangle$ (b). Oscillatory averages cover complete periods. Highlighted markers identify Figs. 4–5. Gray diamonds denote density cycles with unreliable phase near a weak component; their phase-drift values are omitted. Density oscillations disappear between the sampled 6.84 and 6.86 μeV points. Connecting lines guide the eye.

4 TE–TM splitting and spatial polarization

4.1 Extension of the field equations

The two circular fields already span the lower-polariton polarization space. TE and TM are linear combinations with axes set by the in-plane propagation direction; no additional wave functions are required. Separate photon/exciton amplitudes and the upper branch are not resolved.

The field equations become

$$i\hbar\partial_t\psi_\pm = \left[-K\nabla^2 + V_\pm + \frac{i\hbar}{2}(Rn_{A,\pm} - \gamma_C)\right]\psi_\pm - J\psi_\mp - \beta(\partial_x \mp i\partial_y)^2\psi_\mp. \tag{15}$$

The reservoir equations remain those of Eq. (5). The added operator is the TE–TM coupling used in Ref. [6], written in our circular-basis convention. It couples polarization and spatial variation. In Fourier space, its off-diagonal entries are

$$\beta(k_x \mp ik_y)^2 = \beta k^2 e^{\mp 2i\varphi_k}. \tag{16}$$

At $J = 0$, the two linear dispersions have curvatures $K + \beta$ and $K - \beta$, so the TE–TM energy difference has magnitude $2|\beta|k^2$. The mean curvature K is unchanged. The TE–TM term vanishes for a spatially uniform field, but need not vanish at the real-space center of an inhomogeneous condensate: $\mathbf{r} = 0$ is not $\mathbf{k} = 0$.

The three coefficients have distinct roles. The nonlinear term $\alpha_2 n_\mp$ changes a diagonal energy shift; J mixes the components independently of momentum; and β produces direction-dependent mixing. The coherent linear off-diagonal element is

$$b(\mathbf{k}) = -J + \beta k^2 e^{-2i\varphi_k}, \qquad |b|^2 = J^2 + \beta^2 k^4 - 2J\beta k^2 \cos 2\varphi_k. \tag{17}$$

For equal diagonal energies, the splitting $2|b|$ explains the angular dependence of the combined linear coupling. It is not the full excitation spectrum of the pumped nonlinear state.

4.2 TE–TM coupling under equal pumping

We first set $\epsilon = 0$, $J = 0$, and $\beta = \beta_{\mathrm{ref}} = 0.0119$ meV μm^2. The condensate starts with equal real fields. The pump supplies populations through the reservoirs and does not impose a coherent optical phase on the condensate.

Figures 7 and 8 show the total density and the local degree of circular polarization,

$$S_0 = n_+ + n_-, \qquad s_3 = \frac{n_+ - n_-}{n_+ + n_-}. \tag{18}$$

The spot develops four sectors of alternating circular polarization, much clearer in s_3 than in total density. This is consistent with Eq. (16), without assigning a topological charge to the texture.

For the annular pump, component densities have settled by 6000 ps, whereas polarization still changes slowly. This illustrates why density stationarity does not establish polarization stationarity: s_1 and s_2 also depend on relative phase (Chapter 5). A strong normalized signal in a dim region need not imply intense emission; density masks are applied throughout.

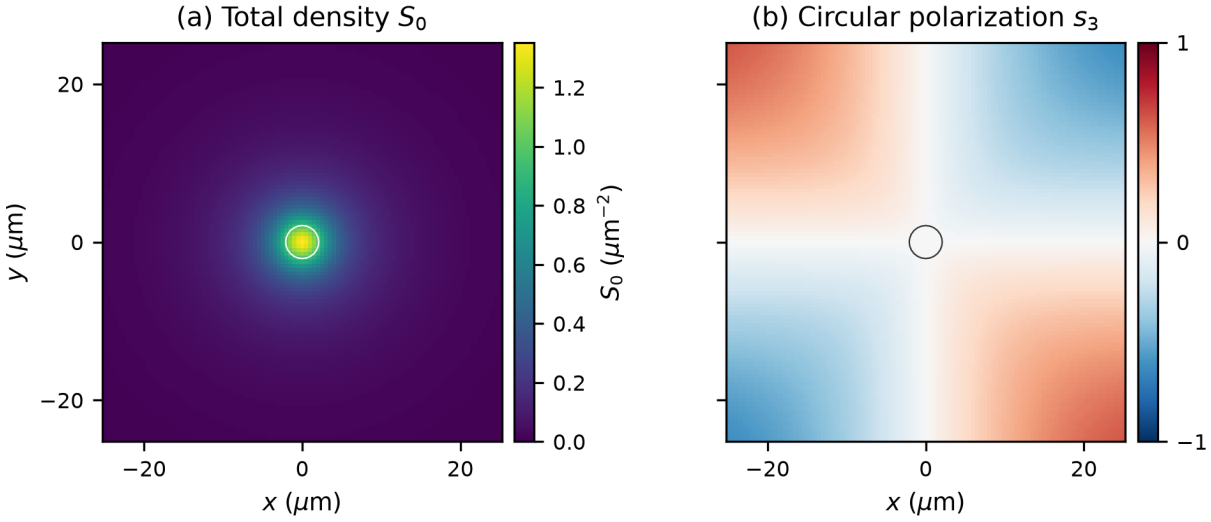


Figure 7: Total density S_0 (a) and local circular polarization s_3 (b) of the Gaussian spot, with $p = 13.8$, $\epsilon = 0$, $J = 0$, and $\beta = 0.0119$ meV μm^2, on 512^2. The late-time polarization has four alternating sectors. The contour marks half the pump maximum; pixels with $S_0 < 10^{-3} \max S_0$ are masked in the polarization map. Sampling details are listed in Table 1.

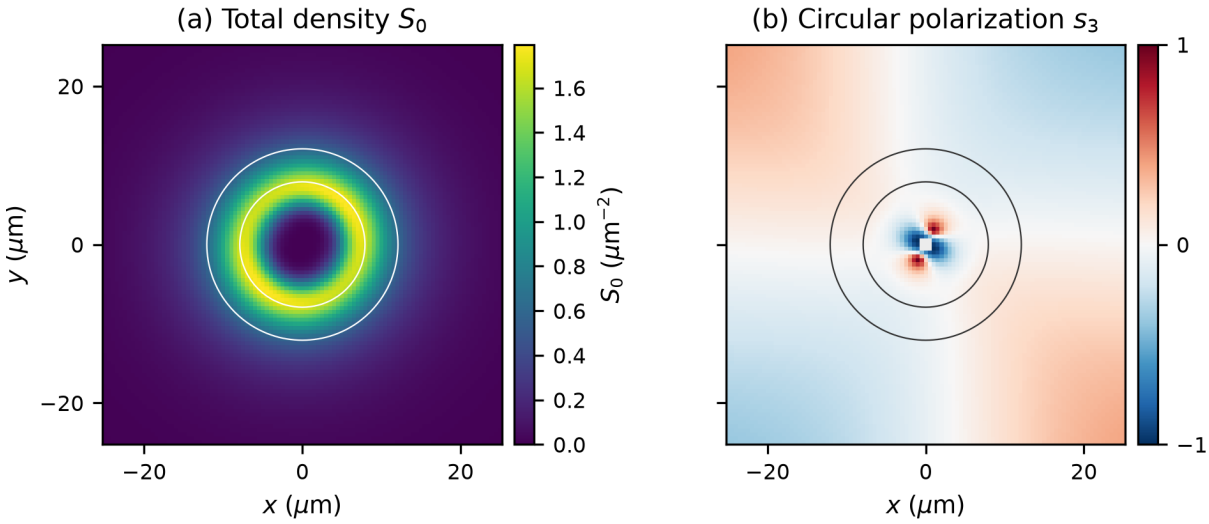


Figure 8: Annular-pump density S_0 (a) and circular polarization s_3 (b) at 6000 ps, with $p = 3.3$, $\epsilon = 0$, $J = 0$, and $\beta = 0.0119$ meV μm^2, on 512^2. Densities have settled, but polarization still evolves. Circles mark half the pump maximum; the polarization mask is defined as in Fig. 7. Sampling details are listed in Table 1.

To quantify the spot's angular response, define

$$C_2(t) = \frac{\int_{\mathcal{R}} S_3(r,\varphi,t) e^{-2i\varphi}\, d^2r}{\int_{\mathcal{R}} S_0(r,\varphi,t)\, d^2r}, \qquad Q_2(t) = 2|C_2(t)|, \qquad S_3 = n_+ - n_-. \tag{19}$$

All coefficients use the analysis annulus $\mathcal{R} = \{5 \le r \le 15\ \mu\text{m}\}$ and the same quadrature. The final-400-ps means of Q_2 are approximately 0.0768, 0.1529, and 0.3008 for $\beta/\beta_{\text{ref}} = 0.5, 1, 2$, with zero for the symmetric uncoupled control (Fig. 9). This nearly linear dependence describes the second harmonic over the sampled range. The varied coefficients test model sensitivity, not experimental uncertainty.

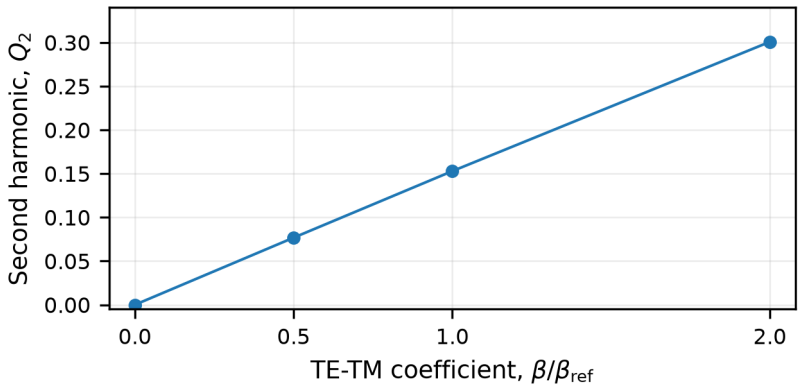


Figure 9: Second angular harmonic $Q_2 = 2|C_2|$, defined in Eq. (19), versus TE–TM strength for the Gaussian spot. All runs use $p = 13.8$, $\epsilon = 0$, $J = 0$, 512^2, and the analysis annulus $5 \le r \le 15$ μm with common quadrature. Each point is the mean of $Q_2(t)$ over 2600–3000 ps, not the modulus of averaged C_2. Bars show the temporal range, not numerical uncertainty. Here $\beta_{\text{ref}} = 0.0119$ meV μm^2; lines guide the eye. Sampling details are listed in Table 1.

4.3 Combined coupling in the annular condensate

For the ring at $\epsilon = 0.3$, we compare no coherent coupling, TE–TM alone, J alone, and both together. Nonzero values are $J = 10$ μeV and $\beta = \beta_{\mathrm{ref}}$; other parameters and preparation are unchanged.

Figure 10 shows

$$s_3(\varphi, t) = \frac{\int_{r_1}^{r_2} S_3(r, \varphi, t) r \, dr}{\int_{r_1}^{r_2} S_0(r, \varphi, t) r \, dr}. \tag{20}$$

This is a density-weighted radial polarization profile, not an unweighted average of local s_3. All cases use $r_1 = 5$ μm and $r_2 = 7.5$ μm.

The controls without TE–TM are nearly angle-independent. TE–TM alone gives four maxima per revolution; combined coupling gives two and changes the mean polarization. The former profile is an instantaneous sample of periodic dynamics, whereas the combined state is stationary within tolerance. These conclusions concern the selected states, not every preparation or time.

The combined-coupling reference has independent time-step and grid checks. Halving the step changes the late-time ring density maps by at most 0.001217%, with a maximum RMS difference of 4.44×10^{-7} in normalized Stokes parameters. The corresponding spot reference has differences of 0.04124% and 8.25×10^{-4}, and its full-map period changes by approximately 0.00355%. These comparisons quantify the tested reference calculations; they are not error bars for every angular profile or for the high-J scan below. Appendix B describes the common-grid comparisons and domain checks.

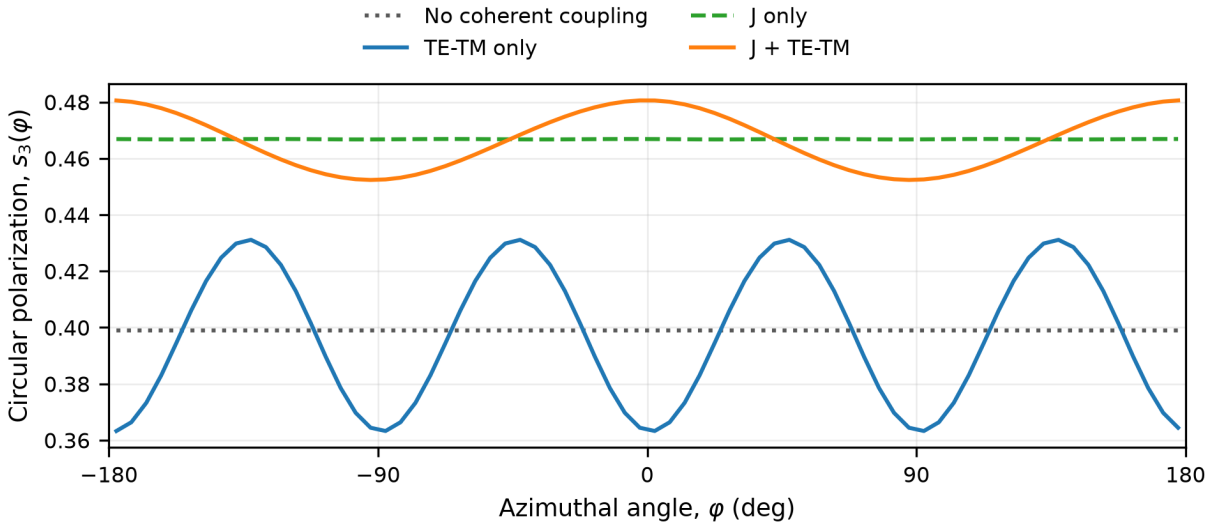


Figure 10: Radially weighted circular-polarization profiles, Eq. (20), for the annular condensate with neither coherent coupling, TE–TM alone, J alone, and both. Nonzero values are $J = 10$ μeV and $\beta = 0.0119$ meV μm^2; $p = 3.3$, $\epsilon = 0.3$, and the 512^2 grid are common. All profiles use $5 \le r \le 7.5$ μm and a common 72-sector polar Gauss quadrature. The sampling times differ between runs (Table 1); no common-time convergence is implied. The TE–TM-only curve is an instantaneous periodic-state profile; the combined state is stationary. Sampling details are listed in Table 1.

4.4 Three dynamical regimes of the Gaussian spot

The spatial scan varies J at fixed $p = 13.8$, $\epsilon = 0.3$, and $\beta = \beta_{\mathrm{ref}}$. Every run starts from the same prescribed seed; neighboring values of J are not obtained by continuing the previous final state.

The integrated observables are

$$A_Z = \frac{Z_{\max} - Z_{\min}}{2}, \qquad \Theta(t) = \arg \int \psi_+(\mathbf{r}, t)\psi_-^*(\mathbf{r}, t)\, d^2r, \qquad \Omega_\Theta = \langle \dot{\Theta} \rangle. \tag{21}$$

Unwrapping requires nonvanishing coherence. Since spatial cancellation can separate Θ from local phase behavior, classification also uses the center and $(\pm 2, 0)$, $(0, \pm 2)$ μm, together with density and Stokes-map recurrence.

The selected examples are $J = 40$, 65, and 100 μeV: running phase, bounded phase oscillations, and fixed phase, respectively (Figs. 11–12). The additional 45 μeV point has a running phase, 50 μeV remains unresolved, and 70 μeV has bounded phase oscillations. The confirmed samples place the running-to-bounded change between 45 and 60 μeV, with unresolved points inside that interval, and the bounded-to-fixed change between 70 and 75 μeV. These are sampled brackets for this preparation, not precise or grid-converged bifurcation values. At 55 μeV, periodic maps coexist with ambiguous local phase near deep component minima, so the phase classification remains unresolved. The 75 μeV state required extension to 4500 ps before stationarity was accepted.

The scan and the three central examples use 256^2, $L = 256$ μm, and $\Delta t = 0.025$ ps in double precision. Selected scan points have finer-grid, smaller-step, and larger-domain controls; the 40, 65, and 100 μeV examples do not all have independent 512^2 checks. Their observed regimes are reported at this resolution, without assigning grid-converged transition values. The stronger couplings than in 0D cannot be attributed to TE–TM alone: pump strength, transport, and reservoir profiles also differ.

Figure 12 shows central imbalance and unwrapped relative phase. The local polarization periods, denoted consistently by T_c, are 34.6402 ps at 40 μeV and 25.0861 ps at 65 μeV; the 100 μeV example is stationary. These values are confirmed local return times, not proven minimal periods or uncertainty bounds. They are half the periods selected by the earlier full-map analyzer. Local Stokes return errors are approximately 2.74×10^{-4} and 1.53×10^{-6}, respectively, and the temporal-interpolation cross-check also passes. Central trajectories and full spatial maps remain different observables; a local return does not establish recurrence of the entire pattern.

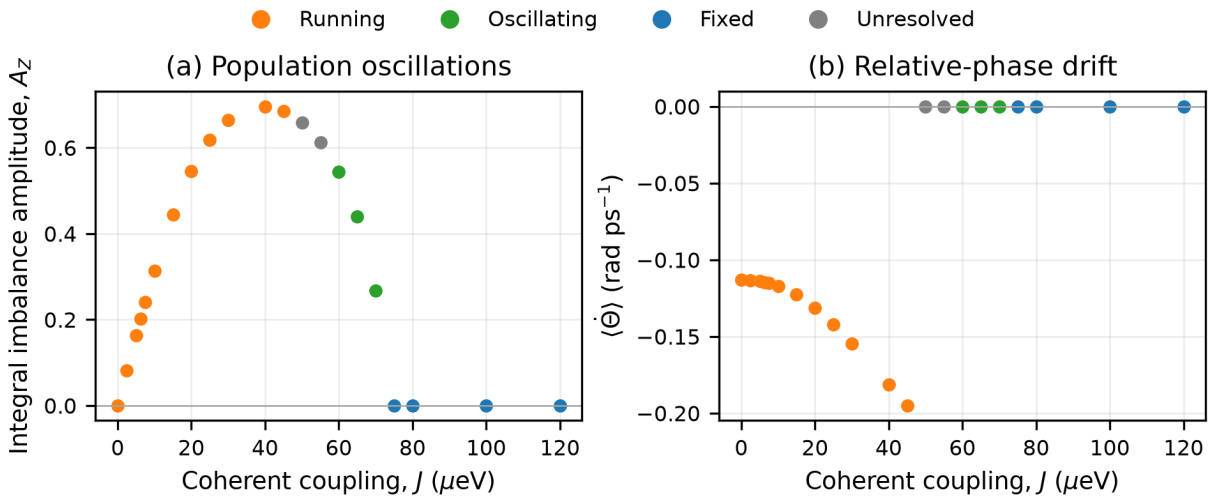


Figure 11: Gaussian-spot coupling scan at $\beta = 0.0119$ meV μm^2, $p = 13.8$, and $\epsilon = 0.3$: integrated imbalance half-amplitude A_Z (a) and integrated-coherence phase drift Ω_Θ (b), defined in Eq. (21). Colors distinguish running, bounded oscillating, fixed, and unresolved phase classifications, with local phase checks. Each point starts independently. The main scan uses 256^2, with selected numerical controls; precise transition boundaries are not established.

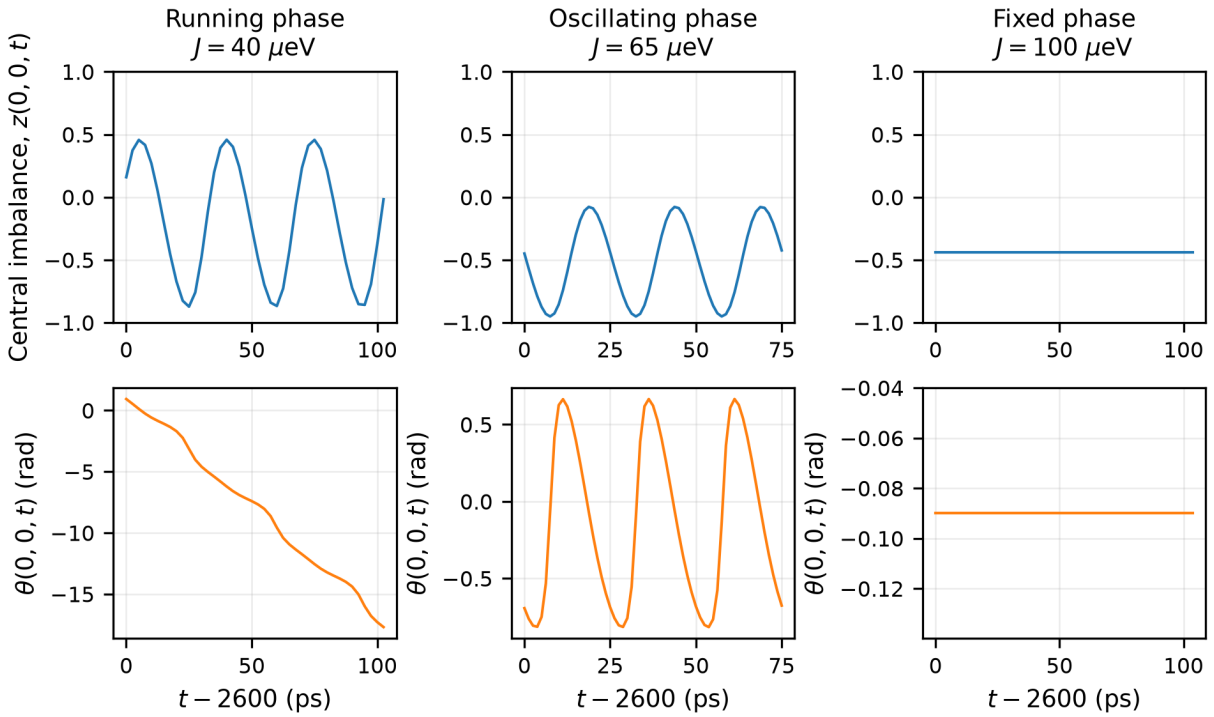


Figure 12: Central dynamics of the 2D Gaussian spot at $J = 40$, 65, and 100 μeV: running phase (a), bounded phase oscillations (b), and fixed phase (c). Top: local imbalance $z(0, 0, t)$. Bottom: unwrapped local phase $\theta(0, 0, t)$. All cases use $p = 13.8$, $\epsilon = 0.3$, $\beta = 0.0119$ meV μm^2, and 256^2. All panels use $t - t_0$, with the common start $t_0 = 2600$ ps. Local polarization periods are approximately 34.6402 and 25.0861 ps in (a,b); the nonzero fixed phase in (c) is retained. Sampling details are listed in Table 1.

5 Stokes parameters and the Poincaré sphere

5.1 Definitions and normalization

The component densities determine circular polarization, but they do not specify the full polarization state. The relative phase is also required. For the circular basis $\mathbf{e}_\pm = (\mathbf{e}_x \pm i\mathbf{e}_y)/\sqrt{2}$, we define

$$\begin{aligned} S_0 &= |\psi_+|^2 + |\psi_-|^2, \\ S_1 &= 2\,\mathrm{Re}(\psi_+^*\psi_-), \qquad S_2 = 2\,\mathrm{Im}(\psi_+^*\psi_-), \\ S_3 &= |\psi_+|^2 - |\psi_-|^2, \qquad s_i = S_i/S_0. \end{aligned} \tag{22}$$

S_i have density units; s_i are dimensionless and normalized by the density at the same position and time, not the image maximum. They describe H/V, ±45°, and circular polarization with signs fixed by the stated basis.

For a coherent two-component field at a point with $S_0 > 0$,

$$\mathbf{s} = \left(\sqrt{1-z^2}\cos\theta, -\sqrt{1-z^2}\sin\theta, z\right), \qquad s_1^2 + s_2^2 + s_3^2 = 1. \tag{23}$$

The instantaneous local state lies on the unit Poincaré sphere: the poles represent circular and the equator linear polarization. The axes label polarization states, not directions of particle motion.

For an unresolved spatial or temporal measurement, average S_i first and divide by averaged S_0. Cancellation can place this vector inside the sphere; it must not be forced back to unit length. The trajectories below instead use instantaneous local fields.

5.2 Spatial polarization maps

Figures 13–14 show simultaneous maps of all three normalized Stokes parameters. Equal component densities imply $s_3 = 0$, while phase variation can still produce structure in s_1, s_2. Round density profiles therefore need not have uniform polarization.

The spot map uses the selected running-phase calculation at $J = 40$ μeV, on 256^2; its center provides the corresponding trajectory in Fig. 16. The ring uses the stationary combined-coupling reference at $J = 10$ μeV, on 512^2. These are illustrations of different states, not a controlled comparison at equal pump and coupling.

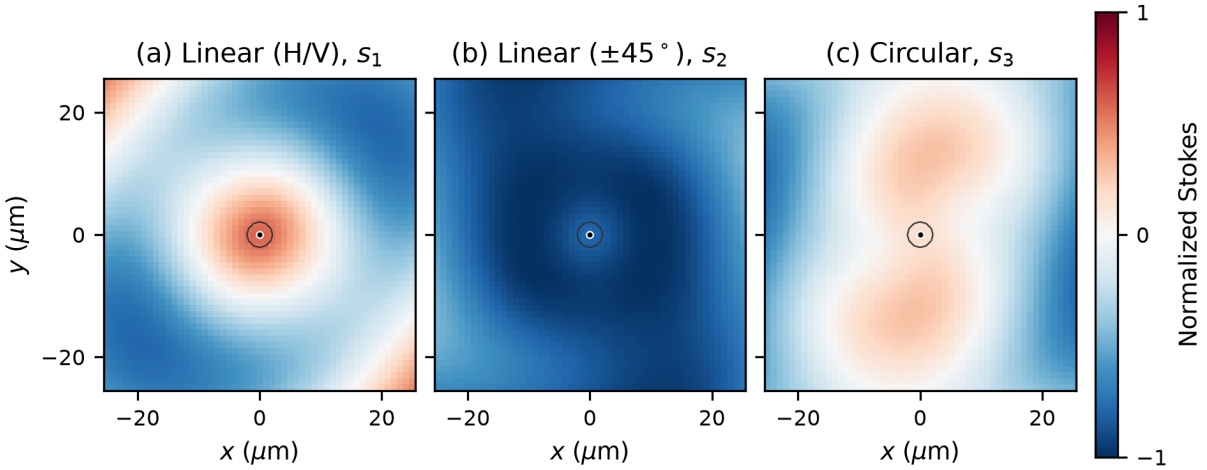


Figure 13: Instantaneous normalized Stokes maps of the running-phase Gaussian spot at $J = 40$ μeV: H/V polarization s_1 (a), $\pm 45^\circ$ polarization s_2 (b), and circular polarization s_3 (c). Parameters and 256^2 grid match Fig. 12(a). Panels show $t = 2600$ ps with the common scale $[-1, 1]$; normalization is local. The contour marks half the pump maximum, and the dot marks the center. The density mask follows Fig. 7. Sampling details are listed in Table 1.

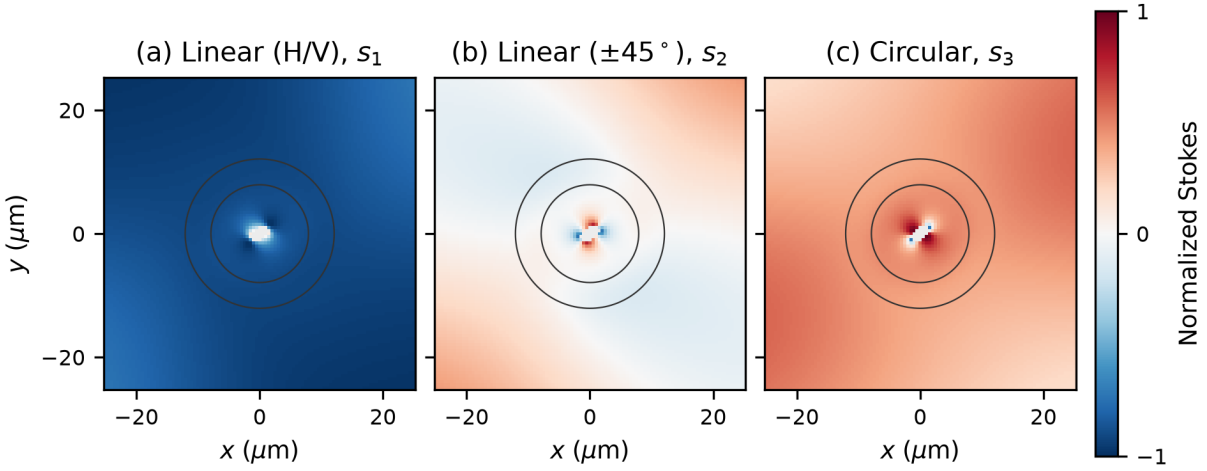


Figure 14: Normalized Stokes maps of the stationary annular reference: s_1 (a), s_2 (b), and s_3 (c), at $J = 10$ μeV, $\beta = 0.0119$ meV μm^2, $p = 3.3$, and $\epsilon = 0.3$, on 512^2. Panels show $t = 3600$ ps with the common scale $[-1, 1]$. Circles mark half the pump maximum; the density mask follows Fig. 7. Sampling details are listed in Table 1.

5.3 The three homogeneous regimes on the sphere

Figure 15 represents the same 0D solutions as Figs. 4–5. Running and bounded phase oscillations both form closed polarization paths; the fixed-phase state is a point.

A closed path does not imply a bounded unwrapped phase: a 2π phase change returns to the same azimuth. Phase traces distinguish these cases. Near a pole, the weak component makes phase ill-conditioned even when the Stokes vector remains well defined.

The fixed point retains the common field rotation and nonzero exchange current discussed in Section 3.2.

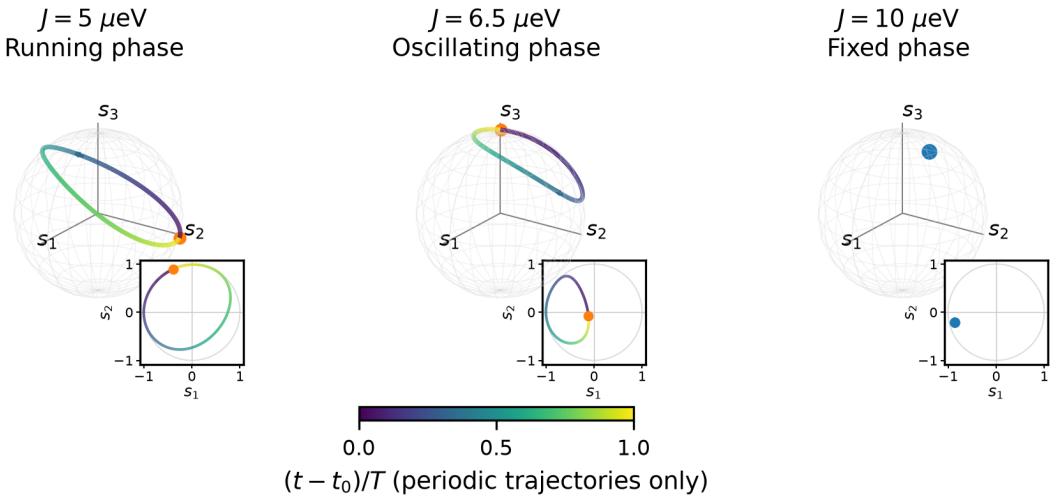


Figure 15: Poincaré-sphere trajectories for the homogeneous states in Figs. 4–5, at $J = 5$, 6.5, and 10 μeV. The oscillatory states show the saved samples spanning approximately one cycle, with measured periods 344.48 and 282.41 ps; the stationary state is a point. Sampling intervals are given in Table 1. Color denotes $(t - t_0)/T$, arrows indicate motion, and a marker identifies each start. Instantaneous fields give $|\mathbf{s}| = 1$, without averaging.

5.4 Polarization dynamics at the center of the 2D spot

Figure 16 uses only $\mathbf{s}(0, 0, t)$ from the 40, 65, and 100 μeV calculations of Fig. 12. All three trajectories start at $t_0 = 2600$ ps. Each oscillatory state is shown over one local period T_c; the stationary state is a point. Sphere axes and viewing angles are identical.

The scan, time traces, and sphere plots thus connect coupling strength, phase classification, and polarization geometry. No additional dynamical approximation is introduced by the sphere representation.

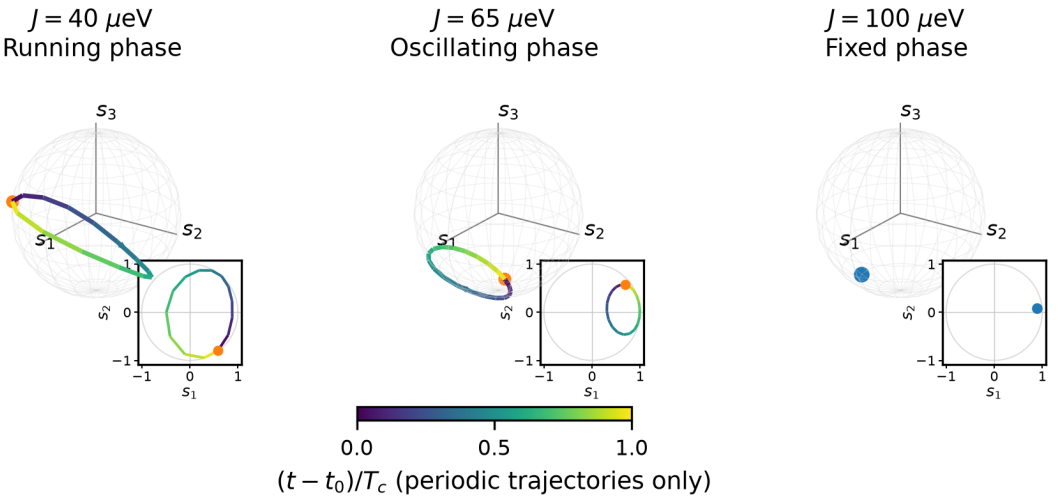


Figure 16: Local Poincaré-sphere trajectories at the Gaussian-spot center for $J = 40$, 65, and 100 μeV, using the same 256^2 calculations as Fig. 12. All cases start at $t_0 = 2600$ ps. The running and bounded-phase states show one local period, $T_c \approx 34.6402$ and 25.0861 ps, respectively; the fixed state is a point. Color denotes $(t - t_0)/T_c$, with arrows and start markers. No spatial averaging is applied; all panels use the same scale and view. Cycle endpoints are interpolated in unnormalized Stokes components and then divided by S_0, without an artificial closing segment or forced unit normalization. Sampling details are listed in Table 1.

Table 1: Snapshot times and analysis settings. Radii are measured from the pump center; times are measured from the simulation start. Populations are integrated over the full computational domain. Rounded values are reported at the times of the displayed snapshots.

Figure	Sampling and analysis settings
3(a,b): spot	$t = 3000$ ps; $N_+ = 116.277$, $N_- = 93.7473$, $Z = 0.107274$.
3(c,d): ring	$t = 6000$ ps; $N_+ = 516.265$, $N_- = 241.631$, $Z = 0.362365$.
7	Spot snapshot at $t = 3000$ ps.
8	Ring snapshot at $t = 6000$ ps.
9	Analysis annulus $5 \leq r \leq 15$ μm; all four runs use $2600 \leq t \leq 3000$ ps.
10	Radial interval $5 \leq r \leq 7.5$ μm. No coherent coupling: 6000 ps; TE–TM only: 6000 ps; J only: 3600 ps; J + TE–TM: 3600 ps.
12	All central traces start at $t_0 = 2600$ ps. Each oscillatory example spans three confirmed local periods; the fixed-state example uses a comparable duration.
13	Spot Stokes maps at $t = 2600$ ps, $J = 40$ μeV: the initial snapshot of the corresponding central trajectory.
14	Ring Stokes maps at $t = 3600$ ps, $J = 10$ μeV.
15	Homogeneous samples: 17000–17344 ps at 5 μeV and 17000–17282 ps at 6.5 μeV, sampled every 0.5 ps; the 10 μeV fixed-state point is taken at 18000 ps.
16	One central cycle: 2600–2634.6402 ps at 40 μeV and 2600–2625.0861 ps at 65 μeV. The 100 μeV fixed state is represented by its point at 2600 ps.

Conclusions

Matching the active–inactive cascade to the earlier scalar model preserves its stationary solutions and homogeneous threshold while changing transients. At $P_A = 1.5P_{\mathrm{th}}$, the half-density arrival is delayed from 159.1 to 188.0 ps, and decay after pump removal is slower.

Reservoir spin relaxation reduces the pump-induced imbalance; the homogeneous steady densities agree with the analytical solution. Constant coherent coupling produces three distinct regimes: a running relative phase, bounded phase oscillations, and a fixed phase. Cessation of phase accumulation and disappearance of density oscillations occur at different couplings.

TE–TM coupling creates spatial circular-polarization structure under equal pumping. For the spot, the second angular harmonic grows approximately linearly across the sampled coupling range. In the annular comparison, adding constant coupling changes the displayed angular profile from four maxima to two. The three dynamical regimes also appear in the 2D spot, illustrated at 40, 65, and 100 μeV. Stokes maps and central sphere trajectories connect these regimes to the polarization state.

The numerical evidence has different scope: the main spatial comparisons use 512^2, with independent controls for the combined-coupling references, whereas the selected high-J examples use 256^2. Their transition boundaries are not yet grid-converged. The equal-pump TE–TM ring still shows slow polarization evolution at 6000 ps. Finally, additional coefficients are literature-scale model choices rather than a common experimental calibration; the results establish behavior of this specified model and preparation.

A Parameters and their relation to the previous reports

A.1 Parameters retained from the condensation model

The base coefficients are retained from the previous report, which used the parametrization of Alyatkin et al. [1]. In particular, R is a model parameter and $g_R = 2g_C$ is an assumption of that model. Retaining these values makes the comparison consistent; it does not imply that every coefficient was independently measured for the structure considered here.

Quantity	Value
$\hbar$	0.658212 meV ps
m/m_e	5.63×10^{-5}
$K = \hbar^2/(2m)$	0.676729 meV μm^2
$g_C = \alpha_1$	0.0024 meV μm^2
$g_R = g_A$	0.0048 meV μm^2
R	0.021 μm^2 ps^{-1}
γ_C	0.182 ps^{-1}
$\gamma_R = \gamma_A$	0.050 ps^{-1}
η	9.8 ps

All densities are expressed in μm^{-2} and pump rates in μm^{-2} ps^{-1}. These are lower-polariton parameters. Neither the polariton mass nor the loss rate is re-extracted from the transfer-matrix model of the first report.

A.2 Inactive reservoir

The matching condition $g_I/W = g_R\eta$ fixes the ratio of two new coefficients, not their individual values. The main parameter set additionally assumes $g_I = g_R$, giving $W = 1/\eta = 0.102041$ ps^{-1}.

The inactive-reservoir loss rate is $\gamma_I = 1/550$ ps^{-1}, following the scale in Ref. [2]. Its other parameters are not adopted as a complete set: γ_A is retained from the condensation report, and W is fixed by matching. The resulting values are

$$g_I = 0.0048 \text{ meV}\,\mu\text{m}^2, \qquad \tau_I = 9.62844 \text{ ps}, \qquad P_{\text{in}} = 1.01782 P_A.$$

The 550 ps time describes loss without transfer to the active reservoir. The response of the cascade is governed by the sum $\gamma_I + W$, hence the much shorter τ_I.

A.3 Spin-dependent coefficients and the scalar limit

We retain $\alpha_1 = g_C$ and choose $\alpha_2 = -0.1\alpha_1 = -0.00024$ meV μm^2, using the model ratio in Ref. [4]. The reservoir rates are $\Gamma_{s,A} = \Gamma_{s,I} = 0.1$ ps^{-1}, following the scale used in Ref. [3]. These are parameters of a phenomenological extension, not measurements of the present structure.

For a term $\Gamma_s(n_{-\sigma} - n_\sigma)$, spin exchange alone makes the population difference decay at $2\Gamma_s$, corresponding to 5 ps for this choice. This factor must be included when comparing relaxation-time conventions.

The spinor model reduces exactly to the scalar two-reservoir model when $J = \beta = \Gamma_{s,A} = \Gamma_{s,I} = 0$, the minus pump is zero, and all minus fields and densities initially vanish. The

zero minus component is then preserved. This limit differs from symmetric pumping of two occupied components. The stationary matching of Eq. (3) applies to the scalar models and does not equate their arbitrary transients.

A.4 Constant coherent coupling

The reference $J = 0.010$ meV corresponds to a linear polarization splitting $2J = 20$ μeV for equal diagonal energies. Splittings of approximately 15–20 μeV were reported by Paraïso et al. [5] for another microcavity. This provides a reference energy scale, not a calibration of the present sample.

The homogeneous and spatial scans vary J independently of the other coefficients. The larger values explored in the spatial scan are a model study; their entire range is not justified as a measured splitting by the 15–20 μeV reference.

A.5 TE–TM coefficient

The new coefficient is taken from the lower-polariton model of Cilibrizzi et al. [6]:

$$\beta_{\mathrm{ref}} = \frac{\Delta_{LT}}{k_{LT}^2} = 11.9\ \mu\mathrm{eV}\,\mu\mathrm{m}^2 = 0.0119\ \mathrm{meV}\,\mu\mathrm{m}^2.$$

In that convention, Δ_{LT} is half the TE–TM splitting at k_{LT}; no additional division by two is needed. The coefficient already belongs to a polariton equation, so no extra photon-fraction factor is applied. It is adopted as a literature scale, not as a fit to our cavity.

The retained curvature satisfies $K > |\beta|$ throughout the chosen range. For β_{ref}, the splitting at $k = 1$ $\mu\mathrm{m}^{-1}$ is 23.8 μeV. The sensitivity calculations use 0, $0.5\beta_{\mathrm{ref}}$, β_{ref}, and $2\beta_{\mathrm{ref}}$.

For calibration to a particular structure, the lower-polariton TE and TM dispersions would have to be calculated for that same structure and detuning, then fitted at small k by their quadratic difference. The present extension does not perform that additional transfer-matrix calculation. In particular, the 2×2 photon–exciton matrix in the dispersion report and the 2×2 circular-polarization operator here act on different degrees of freedom.

B Numerical method and validation

B.1 Homogeneous integration and initial states

In the homogeneous calculations, spatial derivatives are omitted, but two-dimensional density units are retained. The fields and reservoirs are integrated jointly with DOP853, with relative and absolute tolerances 2×10^{-11} and 2×10^{-13}. The retained 0D series starts from

$$\psi_{\pm}(0) = \sqrt{5 \times 10^{-5}}\, e^{\pm 0.1i}, \qquad n_{A,\pm}(0) = n_{I,\pm}(0) = 0.$$

Steady states at $J = 0$ are checked against Eq. (7). Periodic candidates require recurring densities and polarization, stable periods, and persistent amplitudes on extended windows. Slow cases are integrated longer. Figures 4–6 retain the same original 0D series and initial conditions.

The spatial spot starts from $\psi_{\pm}(\mathbf{r}, 0) = \sqrt{10^{-4} f(r)/2}$, with zero relative phase and empty reservoirs. Annular runs multiply both components by the same smooth real perturbation of amplitude 10^{-3}, evaluated continuously on each grid. This imposes no circulation and replaces the earlier 24-plane-wave seed.

B.2 Grid and Fourier step

The main spatial comparisons use a periodic 256×256 μm domain, a 512×512 grid, and $\Delta t = 0.025$ ps. Two complex condensate fields and four real reservoir densities are evolved in double precision; $n_\pm$ are evaluated from the fields rather than integrated as independent variables.

For a square periodic domain, $k_x = 2\pi m_x/L$ and $k_y = 2\pi m_y/L$, where m_x, m_y are the signed discrete Fourier indices. With $\widetilde{\psi}(\mathbf{k}) = \int \psi(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{r}}\, d^2r$, spatial differentiation becomes multiplication by ik_x or ik_y. The kinetic and TE–TM operator is

$$H_F = Kk^2\mathbb{I} + B(\mathbf{k}), \qquad B(\mathbf{k}) = \beta \begin{pmatrix} 0 & (k_x - ik_y)^2 \\ (k_x + ik_y)^2 & 0 \end{pmatrix}. \tag{24}$$

Since $B^2 = \beta^2 k^4 \mathbb{I}$, the exact Fourier propagator over τ is

$$U_F(\tau) = e^{-iKk^2\tau/\hbar} \left[\cos u\, \mathbb{I} - i\frac{\tau}{\hbar}\,\mathrm{sinc}(u) B\right], \qquad u = \frac{\beta k^2 \tau}{\hbar}, \qquad \mathrm{sinc}(u) = \frac{\sin u}{u}. \tag{25}$$

The continuous value $\mathrm{sinc}(0) = 1$ is used at zero argument. For $\beta = 0$, this reduces to the previous scalar kinetic multiplier applied separately to each component.

Each time step consists of a Fourier half-step, a joint local fourth-order Runge–Kutta step, and a second Fourier half-step. The local field equations are

$$\partial_t \psi_\sigma = \left[\frac{R n_{A,\sigma} - \gamma_C}{2} - \frac{iV_\sigma}{\hbar}\right] \psi_\sigma + \frac{iJ}{\hbar}\psi_{-\sigma}, \tag{26}$$

together with the four reservoir equations. All quantities are updated jointly at each Runge–Kutta stage. This real-space operation is not an exact exponential with frozen densities. The combined splitting scheme is second-order accurate in time. The constant coupling J is included in this local step only, not again in Eq. (24).

The propagator can be checked independently through the $\beta = 0$ limit, preservation of the spinor norm during the isolated Fourier step, and the plane-wave eigenenergies $(K \pm \beta)k^2$. For an initially pure + plane wave evolved by this operator alone, the minus population fraction is $\sin^2(\beta k^2 t/\hbar)$.

B.3 Dynamical classification and spatial controls

Population stationarity is checked separately for each component and reservoir. The criteria include less than 0.5% relative variation of each population and less than 1% changes in the corresponding density maps over consecutive late windows. A stationary polarization additionally requires stable Stokes maps. Constant total population alone is insufficient.

For periodic candidates, each of two consecutive analysis windows contains at least five periods and is at least 200 ps long. The period, mean values, amplitudes, and map recurrence must agree within the stated tolerances. The comparison includes both condensate densities, all four reservoirs, and the polarization maps. Nearly vanishing spatial integrals of linear Stokes components are not used to reject a cycle whose local maps remain periodic. If the available history or sampling does not resolve the test, the classification remains unresolved.

The independent half-step and 768^2 tests of the $J = 10$ μeV, $\beta = \beta_{\mathrm{ref}}$ references are quantified in Section 4.3. Comparisons use common spatial coordinates; the periodic spot allows one recorded common cycle shift. Its full-map recurrence period is approximately 53.68 ps. The ring is stationary. These tests do not constitute a completed 1024^2 calculation or a strict error bound.

The larger-domain test supports the combined-coupling reference within the adopted tolerances. It does not validate every scan point. The 256^2 scan has selected grid, step, and domain controls; its high-J central examples remain less thoroughly checked than the 512^2 references. Local-cycle periods reported in Section 4.4 are not substituted for full-map recurrence tests.

Angular observables use a common spatial quadrature. This separates changes in the propagated fields from changes caused by discretizing the analysis region. Polarization maps are masked where $S_0 < 10^{-3} \max S_0$, using the maximum over the full domain. Relative-phase analysis requires a separate check that neither component becomes too weak; a well-defined total density alone does not guarantee a reliable relative phase.

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